

Multiple Choice (10 marks)

1. Two xylem plant cell types that provide support and conduct water and minerals are the
 - (a) parenchyma and companion cells
 - (b) tracheids and vessel elements
 - (c) parenchyma and collenchyma
 - (d) sieve tube members and companion cells
 - (e) collenchyma and vessel elements
2. Hemoglobin is a molecule that binds to both O_2 and CO_2 . There is an allosteric relationship between the concentrations of O_2 and CO_2 . Hemoglobin's affinity for O_2
 - (a) increases as H^+ concentration increases
 - (b) decreases as blood pH decreases
 - (c) decreases in resting muscle tissue
 - (d) increases as blood pH decreases
 - (e) increases as O_2 concentration decreases
3. Explain the mechanism of the genetic determination of sex in man .
 - (a) The temperature during embryonic development determines the sex.
 - (b) The presence of an X or Y chromosome in the mother determines the sex of the offspring.
 - (c) The presence of the X chromosome determines the sex.
 - (d) The sex is determined by environmental factors after birth.
 - (e) The presence of two Y chromosomes determines that an individual will be male.
4. What unusual feature of the cycads makes them a remarkable group among the seed plants?
 - (a) They have flowers that bloom for several years before producing seeds
 - (b) They have a unique leaf structure
 - (c) They can survive in extremely cold climates
 - (d) They produce large seeds
 - (e) Their seeds can remain dormant for up to a century before germinating
5. The energy given up by electrons as they move through the electron transport chain is used to

- (a) produce ATP
- (b) create carbon dioxide
- (c) make FADH 2
- (d) synthesize proteins
- (e) make glucose

True / False (10 marks)

Answer True or False

- 6. True or False: Is the advanced age a contraindication to GERD laparoscopic surgery?
- 7. True or False: Should ascitis volume and anthropometric measurements be estimated in hospitalized alcoholic cirrotics?
- 8. True or False: Perioperative care in an animal model for training in abdominal surgery: is it necessary a preoperative fasting?
- 9. True or False: Does Sensation Return to the Nasal Tip After Microfat Grafting?
- 10. True or False: Is fetal anatomic assessment on follow-up antepartum sonograms clinically useful?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the fundamental differences between prokaryotic and eukaryotic cells, including their structures, functions, and implications for biological processes.
- 12. Discuss the essential characteristics that define living organisms and explain why mere chemical components, even in proper proportions, do not constitute life.

End of Paper